

Manufacturing process Week-2

1. Molding Sand-Types
2. Ingredients and Properties of Molding Sand
3. Importance of Core and Core prints-Gates-Runner-Riser

Moulding Sand

Moulding sand is a specially formulated material used in foundry operations to create moulds and cores for metal casting. It typically consists of silica sand mixed with additives like clay, water, and bonding agents to impart specific properties such as cohesiveness, permeability, and refractoriness.

Types of Moulding Sand

According to their usage, moulding sand can be classified as follows:

1.Green Sand

The green sand contains natural sand with sufficient moisture, comprising a mixture of silica and 15 to 30% clay, along with about 8% water. Clay and water act as bonding materials, giving strength to the mould. Green sand moulds are suitable for simple and rough casting products, used for both ferrous and non-ferrous metals.

2.Dry Sand

When moisture is removed from green sand, it becomes dry sand. Moulds produced with dry sand exhibit greater strength, rigidity, and thermal stability, making it ideal for large and heavy castings.

3.Loam Sand

Loam sand is a blend of 50% sand and 50% clay, with the addition of sufficient water. It is used for crafting large and heavy moulds, such as turbine parts and hoppers.

4.Facing Sand

Facing sand, used for the mould's surface, consists of silica sand and clay without any used sand. It directly contacts the hot molten metal, requiring high refractoriness and strength with very fine grains.



5. Parting Sand

Pure silica sand applied on the pattern's surface before moulding is known as parting sand, preventing the moulding sand from sticking to the pattern during withdrawal. It is also sprinkled on the contact surface of cope, drag, and cheek.

6. Backing or Floor Sand

Backing sand is old and repeatedly used sand with a black colour. It supports the facing sand and fills the entire volume of the mould. This sand accumulates on the floor after casting and is thus called floor sand.

7. Core Sand

Core sand, used for making cores, is a mixture of silica sand with core oil (linseed oil, resin, mineral oil) and other binding materials like Dextrine, corn flour, and sodium silicate, giving it remarkable compressive strength.

Properties of Molding Sand

The important properties of moulding sand are as follows:

1. Porosity
2. Flowability
3. Collapsibility
4. Adhesiveness
5. Cohesiveness or strength
6. Refractoriness

1. Porosity or Permeability:

It is the ability of sand by which it allows the gases to pass through it easily.

Some gases get dissolved in molten metal and when this molten metal starts to solidify, these dissolved gases come out of the molten metal and try to escape out of the moulding sand. If the sand is not enough porous then these gases will not be able to go out of the mould and get trapped into the casting and produce holes and pores in metal casting.

2. Flowability:

The ability of moulding sand to behave like a fluid when it is rammed is called flowability.

Due to this property the sand can easily occupy the space in moulding box and take up its shape. This allows the sand to compress to a compact density and let it pack around the pattern. The sand should be of high flowability, so that it can be easily compacted for uniform density and to obtain a good impression of the pattern in the mould. The flowability of the sand can be increased as we increase the clay and water content in the sand.



3. Collapsibility:

The ability of the moulding sand to collapse after solidification of the molten metal is called collapsibility.

After the solidification of molten metal, the sand should get collapse for free contraction of the metal. If free contraction of the metal will happen then it eliminates naturally the tearing or cracking of the contracting metal.

4. Adhesiveness:

The ability of the sand particles to get stick with another body is called adhesiveness.

The sand should have sufficient adhesiveness so that it can easily get cling to the sides of the moulding boxes and does not fall out to the box when it is removed.

5. Cohesiveness or Strength

The ability of the sand particles to stick with each other is called cohesiveness.

The strength of the sand depends upon how cohesive the sand particles are. The sand should have sufficient strength so that it can easily capable to retain its shape during conveying, turning or closing and pouring.

6. Refractoriness

The ability of the moulding sand to withstand the high temperature of the molten metal without fusing into it is called refractoriness.

The moulding sand must have enough refractoriness property to produce excellent quality of casting free from defects. The sand with lack of refractoriness melts and gets fuse in the casting and spoils the quality of the cast metal. The refractoriness is the measure of sinter point of the sand not its melting point.

Constituents of Moulding Sand

The various constituents of Moulding sand are:

Silica Sand: The main component providing strength, stability, and permeability, with minor impurities like iron oxide, alumina, limestone, magnesla, soda, and potash.

Binder: Inorganic binders like clay, sodium silicate, and Portland cement, or organic binders like dextrin, molasses, cereal binders, linseed oil, and resins.

Moisture: Typically 2% to 8%, added to develop bonds and increase strength in the sand.

Additives: Materials added to enhance specific properties of moulding and core sand.

Coal Dust: Creates a reducing atmosphere during casting, mainly used for grey iron and malleable cast iron castings.



Corn Flour: Increases collapsibility and allows free movement of sand grains, reducing mould expansion and defects.

Dextrin: Improves the dry strength of moulds.

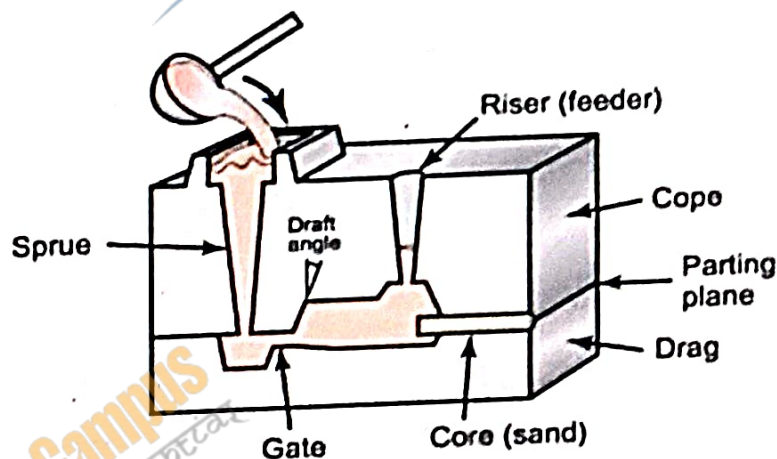
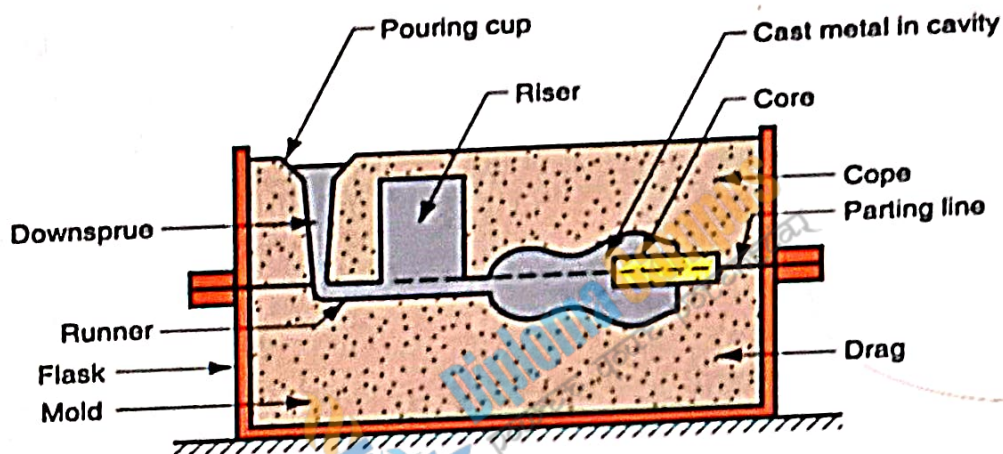
Sea Coal: Fills the pores, reducing mould wall movement and permeability for a clean and smooth mould surface.

Pitch: Enhances hot strength and surface finish.

Wood Flour: Prevents sand grains from making contact, increases mould wall movement, and reduces expansion defects.

Pulverised Silica: Increases hot strength and surface finish, reducing metal penetration in moulds and cores.

CASTING TERMINOLOGY



(1) **Flask:** The Flask is the box that contains the moulding aggregate.

(2) **Pattern:** It is the replica of the final object to be made with some modification.

(3) Parting Line:- This is dividing line b/w two parts of flask that makes the sand moulds.

(4) Bottom board:- This is normally made of wood which is used to support the one end of the flask.

(5) Facing sand:- The Small amount of carbonaceous sand material sprinal on the internal mould surface of cavity to give better surface finish to the casting.

(6) Moulding Sand:- It is Freshly Prepare refractory use for making the mould cavity.

(7) Backing Sand:- It is the bulk of the sand used to back up the facing sand & to fill up the volume of the flask.

(8) Pouring basin:- A Small Funnel Shaped Cavity at the top of mould in to inch the molten metal is poured.

(9) Core:- A core is used in Casting and moulding Process to produce internal hollow cavities in final product.

(10) Sprue:- It is Vertical Cavity in the moulds to flow the molten metal from the pouring basin.

(11) Runner:- The pass way in the parting line through which molten metal regulated before they reach to mould cavity.

(12) Chills:- Chilles are metallic object which are placed in mould to increase the cooling rate of the casting.

(13) Chaplet:- It is used to support core inside the mould cavity to take core of it's own weight & over come meler lostetic Forces.

(14) Riser :- It is reservior of molten metal provided in casting show that metal can flow back into the more cavity there is a reduction in volume of metal due to solidify.

